Three proofs of the Fundamental Theorem of Algebra

YIMING LI

Abstract

This note gives three proofs of the fundamental theorem of algebra 0.0.1, all of which are of different flavors. However, we should point out that all existing proofs necessarily involve certain facts from Analysis (or Topology). Furthermore, the theorem, despite having such a name due to historical reasons, is neither fundamental nor algebraic from a modern perspective; it is indeed a theorem, though.

Theorem 0.0.1. *The field of complex numbers is algebraically closed.*

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1 A proof from Topology

Lemma 1.0.1. *If $A \subseteq X$ is a retract of X. Then the homomorphism of fundamental groups induced by the inclusion $j : A \to X$ is injective.*

Proof. Let $r : X \to A$ be the retraction. Consider the composition $\mathrm{id} = r \circ j$. We have $\mathrm{id}_* = (r \circ j)_* = r_* \circ j_*$. So j_* is injective. □

Corollary 1.0.2. *There is no retraction from B^2 to S^1.*

Proof. Use Lemma 1.0.1. □

Proposition 1.0.3. *Let $f : X \to Y$ be continuous. If X is compact and Y is Hausdorff, then f is a closed map.*

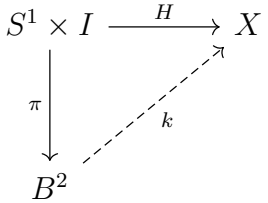
Proof. Take an arbitrary $A \subseteq X$. Since X is compact, so is A. Since continuous maps preserve compactness, $f(A)$ is compact in Y. Since Y is Hausdorff, $f(A)$ is closed, as desired. □

Theorem 1.0.4. *Let $h : S^1 \to X$ be a continuous map. Then TFAE:*

1. *h is nulhomotopic.*
2. *h extends to a continuous map $k : B^2 \to X$.*
3. *h_* is trivial.*

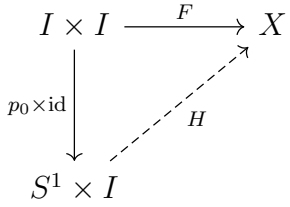
Proof.

(1) $\implies$ (2) : Let $H : S^1 \times I \to X$ be a homotopy between h and a constant map where $h = H(-, 0)$. Define: $\pi : S \times I \to B^2$ by $\pi(x, t) := (1 - t)x$. Then π is continuous, surjective, and by Proposition 1.0.3, closed. Hence π is a quotient map. Then use Theorem 22.2 in [3].



(2) $\implies$ (3) : Let $j : S^1 \to B^2$ be the embedding. Then $h_* = (k \circ j)_* = k_* \circ j_*$. But j_* is trivial, so is h_*.

(3) $\implies$ (1) : Let $p : \mathbb{R} \to S^1$ be the standard covering map and let $p_0 : I \to S^1$ be its restriction to the unit interval. Let $x_0 = h(b_0)$. Since h_* is trivial, the loop $f := h \circ p_0$ represents the identity element in $\pi_1(X, x_0)$. Therefore, there exists a path homotopy between f and the constant path at x_0, say $F : I \times I \to X$. Note that the map $p_0 \times \mathrm{id} : I \times I \to S^1 \times I$ is a quotient map, so by Theorem 22.2 in [3], it induces a continuous map $H : S^1 \times I \to X$. Note that it is a homotopy between h and a constant map.



□

Corollary 1.0.5. *The embedding $j : S^1 \to \mathbb{R}^2 - \mathbf{0}$ is **not** nulhomotopic.*

Proof. There is a retraction: $\mathbb{R}^2 - \mathbf{0} \to S^1$ given by $x \mapsto x/\|x\|$. Hence j_* is injective, hence not trivial. □

Definition 1.0.6. A *vector field* on B^2 is an ordered pair $(x, v(x))$, where $x \in B^2$ and v is a continuous map from B^2 to $\mathbb{R}$. We say that a vector field is *non-vanishing* if $v(x) \neq 0$ for every $x \in B^2$. In that case, v is a continuous map from B^2 to $\mathbb{R} - \mathbf{0}$.

Theorem 1.0.7. *Given a non-vanishing vector field on B^2, there exists a point on S^1 where the vector field points directly inward.*

Proof. Suppose the contrary; then consider the map $v : B^2 \to \mathbb{R} - \mathbf{0}$. Let $w := v|_{S^1}$. Since w extends to v, by Theorem 1.0.4, w is nulhomotopic. On the other hand, we may define:

$$F : S^1 \times I \longrightarrow \mathbb{R} - \mathbf{0}$$

$$(x, t) \longmapsto tx + (1 - t)w(x)$$

F is continuous and $F(\bullet, 0) = w(\bullet), F(\bullet, 1) = \bullet$. Therefore, to show that F is a homotopy between w and the embedding $j : S^1 \to \mathbb{R} - \mathbf{0}$, it suffices to check that $F(x, t)$ never touches $\mathbf{0}$. Clearly $F(x, t) \neq \mathbf{0}$ for $t = 0$ and $t = 1$. Suppose for a contradiction that $F(x_0, t_0) = \mathbf{0}$ for some $t_0 \in (0, 1)$. Then

$$w(x_0) = \frac{t_0}{t_0 - 1} x_0$$

which is a negative scalar of x_0. But that means $w(x_0)$ points directly inward at x_0. Therefore, w is homotopic to j. But by Corollary 1.0.5, j is not nulhomotopic; hence, we run into a contradiction. □

Remark. Applying the same argument to the vector field $(x, -v(x))$, we know that v has to point directly outward at some point on S^1.

We will play the same strategy above to prove Theorem 0.0.1. But first, a beautiful corollary:

Corollary 1.0.8 (Brouwer fixed-point theorem for B^2)**.** *Let $f : B^2 \to B^2$ be continuous; then there exists a point $x \in B^2$ s.t. $f(x) = x$.*

Proof. Suppose the contrary that $f(x) \neq x$ for all $x \in B^2$. Then $(x, f(x) - x)$ is a non-vanishing vector field on B^2. Therefore, by Theorem 1.0.7, there exists a point $x_0 \in S^1$ such that

$$f(x_0) - x_0 = cx_0$$

for some positive c. But that means $f(x_0) = (1 + c)x_0$, i.e. f maps x_0 to somewhere outside B^2, a contradiction. □

Now, we turn to Theorem 0.0.1. But first, some preparation:

Proposition 1.0.9. *Consider $f : S^1 \to S^1$ given by $f(z) = z^n$ for some positive integer n. Then the induced homomorphism f_* is injective.*

Proof. Let $p_0 : I \to S^1$ be the standard loop in S^1. Then its image under f_* is the loop $f(p_0(s)) = (\cos 2\pi ns, \sin 2\pi ns)$. This loop lifts to the path $s \mapsto ns$ in the covering space $\mathbb{R}$. Hence $f \circ p_0$ corresponds $n \in \mathbb{Z}$. f_* is therefore injective. □

Proposition 1.0.10. *Consider $g : S^1 \to \mathbb{R} - \mathbf{0}$ given by $g(z) = z^n$. Then g is not nulhomotopic.*

Proof. Note that $g = j \circ f$ where j is the embedding and f is the map defined in the previous proposition. Then $g_* = j_* \circ f_*$ is injective since both f_* and j_* are injective. Hence g is not nulhomotopic by Theorem 1.0.4. □

Finally, we prove Theorem 0.0.1.

Proof of Theorem 0.0.1. Given a polynomial equation

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$$

We may assume that $|a_{n-1}| + \cdots + |a_0| < 1$. We will show that the above equation has a root lying in B^2. Suppose the contrary. Then we can define a continuous map $k : B^2 \to \mathbb{R} - \mathbf{0}$ by

$$k(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_1z + a_0$$

Note that $k|_{S^1}$ is nulhomotopic by Theorem 1.0.4.

Now we construct a homotopy between $k|_{s^1}(z)$ and $g(z) = z^n$ (restricted to S^1). Once we can do that, by the previous proposition, g is not nulhomotopic and we obtain a contradiction. Let $F(z, t) := tk(z) + (1 - t)z^n = z^n + t(a_{n-1}z^{n-1} + \cdots + a_1z + a_0)$. It remains to check that $F(z, t)$ never touches $\mathbf{0}$ where $t \in (0, 1)$ and $z \in S^1$. This can be seen by

$$\begin{aligned} |F(z,t)| &\geq |z^n| - |t(a_{n-1}z^{n-1} + \cdots + a_1z + a_0)| \\ &\geq 1 - t(|a_{n-1}| + \cdots + |a_0|) \\ &> 0 \end{aligned}$$

□

2 A proof from Complex Analysis

For this section, I basically copied my notes on [2].

Definition 2.0.1. The Laurent series

$$\sum_{n=-\infty}^{\infty} a_n(z-c)^n$$

is called the *Laurent expansion* of $f(z)$ about c. The point c is called an isolated singular point of $f(z)$ and the part of the Laurent expansion with negative exponents:

$$\sum_{n=-\infty}^{-1} a_n(z-c)^n = \cdots + \frac{a_{-2}}{(z-c)^2} + \frac{a_{-1}}{z-c}$$

is called the *principal part* of $f(z)$ at c.

We distinguish the following three cases: (We are actually classifying isolated singularities)

Case 1. There is no principal part. In this case we have for $0 < |z-c| < r(c)$,

$$f(z) = a_0 + a_1(z-c) + a_2(z-c)^2 + \cdots$$

By defining $f(c) = a_0$, we make $f(z)$ into a function which is also holomorphic at c. In this case, c is called a *removable singularity* of $f(z)$. Usually, one removes removable singularities from the start making $f(z)$ into a function which is holomorphic on the whole D.

Let $f(z)$ be holomorphic on D. A point $c \in D$ such that $f(c) = 0$ is called a zero of $f(z)$. Consider the Taylor expansion of $f(z)$ around the zero c:

$$f(z) = a_0 + a_1(z-c) + a^2(z-c)^2 + \cdots$$

Then $a_0 = f(c) = 0$, and it is possible that some of the other coefficients are also equal to 0. If all coefficients $a_1, a_2, ...$ are equal to 0, then $f(z)$ is identically equal to 0 in a neighborhood of c. Let us assume that not all coefficients are equal to 0 and that a_m is the first one that is not equal to 0. Then

$$f(z) = (z-c)^m(a_m + a_{m+1}(z-c) + a_{m+2}(z-c)^2 \cdots)$$

In this case, m is called the *order*, or the *multiplicity of the zero* c. Writing

$$f(z) = (z-c)^m g(z)$$

Then, $g(z)$ is holomorphic on $U_r(c)$ and $g(c) = a_m \neq 0$. Therefore, $g(z) \neq 0$ on a sufficiently small $U_\varepsilon(c)$. Hence

$$f(z) = (z-c)^m g(z) \neq 0 \qquad \text{for } 0 < |z-c| < \varepsilon$$

Thus, we have proved that there are no other zeros in a sufficiently small neighborhood of a zero of a holomorphic function. The zero c is therefore `"isolated"`.

Case 2. The principal part has only a finite number of terms. That is, for $0 < |z-c| < r(0)$, we have

$$f(z) = \frac{a_{-m}}{(z-c)^m} + \cdots + \frac{a_{-1}}{z-c} + \sum_{n=0}^{\infty} a_n (z-c)^n, \qquad a_{-m} \neq 0$$

In this case c is called a *pole* of order m of the function $f(z)$. The function $g(z)$, defined by

$$g(z) = (z-c)^m f(z) = a_{-m} + a_{-m+1}(z-c) + \cdots$$

is holomorphic on $U_{r(c)}(c)$ and $g(c) = a_{-m} \neq 0$. Therefore, $g(z) \neq 0$ on a sufficiently small neighborhood $U_\varepsilon(c)$. From

$$f(z) = \frac{1}{(z-m)^m} g(z)$$

we have

$$\lim_{z \to c} |f(z)| = \frac{|g(z)|}{|z-c|^m} = +\infty$$

In this case, we write $f(c) = +\infty$. We will return to the meaning of ∞ later.
Putting $h(z) = 1/g(z)$ we have

$$\frac{1}{f(z)} = (z-c)^m h(z)$$

and $h(z)$ is holomorphic on $U_\varepsilon(c)$ and $h(z) \neq 0$ for $z \in U_\varepsilon(c)$. Therefore, the pole of the mth-order c of $f(z)$ is a zero of the mth-order of $1/f(z)$. Conversely, it is also true that a zero of the mth order of $1/f(z)$ is a pole of the mth of $f(z)$.

Case 3. The principal part is an infinite series. That is, $a_{-n} \neq 0$ for infinitely many natural numbers n. in this case, c is called an *essential singularity* of $f(z)$. As the following theorem shows, the behavior of $f(z)$ in the neighborhood of an essential singularity is very complicated.

Theorem 2.0.2 (Weierstrass's Theorem)**.** *Let c be an essential singularity of $f(z)$ and w be a complex number. It is possible to find a sequence $\{z_n\}$ of points converging to c such that $\lim_{n\to\infty} f(z_n) = w$. It is also possible to find a sequence $\{z_n\}$ converging to c such that $\lim_{n\to\infty} |f(z_n)| = +\infty$.*

We first prove a proposition:

Proposition 2.0.3. *If c is an isolated singularity of $f(z)$ such that f is bounded on some neighborhood of c, then c is a removable singularity.*

Proof. By assumption there exists a $\delta > 0$ and a M s.t. $|f(z)| \leq M$ for all $|z-c| < \delta$. Recall that

$$a_{-n} = \frac{1}{2\pi i} \int_{\gamma_r} (\zeta - c)^{n-1} f(\zeta) d\zeta$$

Hence

$$|a_{-n}| \leq \frac{1}{2\pi} 2\pi r M = rM$$

This holds for arbitrarily small r, forcing a_{-n} to be zero. □

Proof of Theorem 2.0.2. We first show that $f(z)$ can go to $+\infty$. Suppose the contrary, that is, if there does not exist a sequence $\{z_n\}$ converging to c s.t. $f(z_n) \to +\infty$, then f is bounded in a small neighborhood of c. By our above proposition, c is removable, contradicting our assumption that c is essential.

Then we show that $f(z_n)$ may converge to an arbitrary number with a proper choice of $\{z_n\}$. Fix an arbitrary $w \in \mathbb{C}$. Suppose the contrary that $\nexists$ sequence $\{z_n\}$ converging to c s.t. $f(z_n) \to w$. That means $f(z)$ would eventually "keep a distance" from w no matter which way you try to approach. That is, $\exists\, \varepsilon > 0, \delta > 0$ s.t.

$$|f(z) - w| \geq \varepsilon \qquad \text{whenever } 0 < |z - c| < \delta$$

Now, define

$$g(z) := \frac{1}{f(z) - w}$$

Note that g is holomorphic on $0 < |z - c| < \delta$ and bounded by $1/\varepsilon$. Hence, by our proposition, c is a removable singularity of g. So we can define $g(c)$ so that g is holomorphic on the entire $|z - c| < \delta$. Now we have 2 cases:

1. $g(c) = 0$. Then suppose that c is the $m^{\text{th}}-$order zero of g, i.e., $g(z) = (z - c)^m h(z)$. Then

   $$f(z) = w + \frac{1}{(z - c)^m h(z)}$$

   so c is a $m^{\text{th}}-$order pole of f. Contradiction.

2. $g(c) \neq 0$. Then $1/g$ is holomorphic around c. Now we have

   $$f(z) = w + \frac{1}{g(z)}$$

   Meaning that c is also a removable singularity of f. Contradiction.

□

A function holomorphic on the whole complex plane $\mathbb{C}$ is called an *entire function*. If $f(z)$ is an entire function, then $f(z)$ can be expanded in a power series

$$f(z) = a_0 + a_1 z + a_2 z^2 + \cdots$$

which converges absolutely on the whole complex plane $\mathbb{C}$. We distinguish the following three cases:

1. $f(z) = a_0$
2. $f(z) = a_0 + a_1 z + \cdots + a_n z^n$
3. $f(z) = a_0 + a_1 z + \cdots$

The first case is trivial. For the second case, it is easy to see that $|f(z)| \to +\infty$ as $|z| \to \infty$.

In the third case, $f(z)$ is called a *transcendental entire function*. If $f(z)$ is a transcendental entire function and w an arbitrary complex number, then there exists a sequence $\{z_n\}$ s.t. $\lim_{n\to\infty} |z_n| = \infty$ and $\lim_{n\to\infty} f(z_n) = w$; There also exists a sequence $\{z_n\}$ s.t. $\lim_{n\to\infty} f(z_n) = \infty$ as $|z_n| \to \infty$.

To see this, let $g(\zeta) = f(1/\zeta)$. Then

$$g(\zeta) = \cdots + \frac{a_2}{\zeta^2} + \frac{z_1}{\zeta} + a_0$$

is a holomorphic function of ζ on $\mathbb{C} - \{0\}$. Moreover, 0 is an essential singularity of $g(\zeta)$. So we can apply Theorem 2.0.2 to $g(\zeta)$.

Theorem 2.0.4 (Liouville)**.** *A bounded entire function is a constant function.*

Proof. If f is a bounded entire function, then case 2 and 3 cannot happen. That is, $f(z) = a_0$. □

Proof of Theorem 0.0.1. Let $f(z) = a_0 + a_1 z + \cdots + a_m z^m$ be a non-constant polynomial. Suppose the contrary that f has no zeros in $\mathbb{C}$. Then $1/f(z)$ is an entire function. Moreover, it is bounded. (Fix $R > 0$. For $|z| > R$, $1/f(z)$ is clearly bounded. For $|z| \leq R$, since $1/f$ is continuous, by Heine-Borel, $1/f(z)$ is bounded.) Hence by Liouville, f is a constant. A contradiction. □

3 A proof from (almost) Algebra

Here we say "almost" because we have to admit an elementary result from analysis:

Proposition 3.0.1. *Any polynomial in $\mathbb{R}[x]$ of odd degree has at least one root in $\mathbb{R}$.*

This is a direct corollary of the Intermediate Value Theorem. The main reference of this section is [1].

Let p be a fixed prime. Let's first recall some basic group theory:

Theorem 3.0.2 (Cauchy)**.** *If G is a finite group with $p \mid |G|$. Then G contains an element of order p.*

Definition 3.0.3. A group in which every element has order a power of p is called a *$p-$group*. If H is a subgroup of a group G' and H is a $p-$group, then H is called a *$p-$subgroup* of G'.

Corollary 3.0.4. *A finite group G is a $p-$group if and only if $|G|$ is a power of p.*

Proof. $\Longleftarrow$: trivial. $\Longrightarrow$: Suppose $\exists$ a prime q s.t. $q \mid |G|$. Then by Cauchy, G contains an element of order q. But since G is a $p-$group, every element of G, every element of G has order of a power of p. This implies that $p = q$. Hence $|G|$ is a power of p. □

We shall use the above corollary as the characterization of $p-$groups in our mind.

Theorem 3.0.5 (First Sylow Theorem)**.** *Let G be a group of order $p^n m$ where $n \geq 1$ and $(p, m) = 1$. Then for every $i \in [n]$, G contains a subgroup of order p^i. Moreover, every subgroup of order p^i $(i < n)$ is normal in some subgroup of order p^{i+1}.*

Remark. A subgroup P of a group G is called a *Sylow $p-$subgroup* if P is a maximal $p-$subgroup of G. Theorem 3.0.5 implies that for every prime $p \mid |G|$ (G finite), G has a non-trivial Sylow $p-$group.

Now we go into the proof of our main theorem. But first, some preparation:

Theorem 3.0.6 (Primitive Element Theorem)**.** *Let F be a finite field extension of K. Then*

(1). If F is separable over K, then $F = K(u)$ where u is algebraic over K.

(2). (Artin) More generally, $F = K(u)$ if and only if there are only finitely many intermediate fields.

Lemma 3.0.7. *There is no field extension of dimension 2 over $\mathbb{C}$.*

Proof. Suppose the contrary and let $F \supset \mathbb{C}$ where $[F : \mathbb{C}] = 2$. Then by Theorem 3.0.6 we may assume that $F = \mathbb{C}(u)$ where the corresponding minimal polynomial of u is $f(x) = x^2 + bx + c \in \mathbb{C}[x]$. We know that $f(x) = 0$ has roots of the form $(-b \pm \sqrt{b^2 - 4c})/2$. But every complex number has square roots in $\mathbb{C}$ (because every argument can be divided by 2). So $u \in \mathbb{C}$, a contradiction. □

Proof of Theorem 0.0.1. It suffices to show that $\mathbb{C}$ has no finite dimensional extensions except itself. Since $[\mathbb{C} : \mathbb{R}] = 2$ and $\operatorname{char} \mathbb{R} = 0$, every finite dimensional extension, say $E_1 \supset \mathbb{C}$ is finite and separable over $\mathbb{R}$. Then, By Theorem 3.16 of [1], let F be the normal closure of $\mathbb{R}$, we have $E_1 \subseteq F$. Therefore, in order to show $E_1 = \mathbb{C}$, it suffices to show that $F = \mathbb{C}$.

Note that again by Theorem 3.16 of [1], F is finite dimensional and Galois over $\mathbb{R}$, hence $\mathrm{Aut}_{\mathbb{R}}F$ is finite. We have

$$|\mathrm{Aut}_{\mathbb{R}}F| = |\mathrm{Aut}_{\mathbb{C}}F| \cdot [\mathrm{Aut}_{\mathbb{R}}F : \mathrm{Aut}_{\mathbb{C}}F] = |\mathrm{Aut}_{\mathbb{C}}F| \cdot [\mathbb{C} : \mathbb{R}] = 2|\mathrm{Aut}_{\mathbb{C}}F|$$

So by Theorem 3.0.5 and its remark, $\mathrm{Aut}_{\mathbb{R}}F$ has a $2-$Sylow subgroup of order 2^n for some $n \geq 1$, say H. Let F^H be its fixed field. Note that $[\mathrm{Aut}_{\mathbb{R}}F : H]$ is odd, so is $[F^H : \mathbb{R}]$. Therefore, by Theorem 3.0.6, we may assume that $F^H = \mathbb{R}(u)$ where the corresponding minimal polynomial in $\mathbb{R}[x]$ has odd degree. But by Proposition 3.0.1, this degree has to be 1. Consequently, $u \in \mathbb{R}$ and $F^H = \mathbb{R}$. Hence $H = \mathrm{Aut}_{\mathbb{R}}F$ and so $|\mathrm{Aut}_{\mathbb{C}}F| = 2^{n-1}$. Our goal is to show that $n - 1 = 0$.

Suppose the contrary that $n - 1 > 0$. Then, by Theorem 3.0.5 again, $\mathrm{Aut}_{\mathbb{C}}F$ has a subgroup of index 2, say J. Let F^J be its fixed field. Then $[F^J : \mathbb{C}] = 2$, contradicting to Lemma 3.0.7. So we are done. □

Corollary 3.0.8. *Every proper algebraic extension of* $\mathbb{R}$ *is isomorphic to* $\mathbb{C}$.

Proof. $\mathrm{Aut}_{\mathbb{R}}\mathbb{C}$ has no proper subgroup, hence there is no proper intermediate field between $\mathbb{R}$ and $\mathbb{C}$. There is also no algebraic extension of $\mathbb{R}$ that properly contains $\mathbb{C}$ since $\mathbb{C}$ is algebraically closed. Therefore, the only possible algebraic extension of $\mathbb{R}$ is $\mathbb{C}$, up to isomorphism. □

References

[1] Thomas W. Hungerford. *Algebra*, volume 73 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1974.

[2] Kunihiko Kodaira. *Complex Analysis*. Cambridge University Press, Cambridge, UK, 2007.

[3] James R. Munkres. *Topology*, volume 2. Prentice Hall, Upper Saddle River, NJ, 2000.